

INDEFINITE INTEGRAL

1. If $\int \frac{\cos 4x + 1}{\cot x - \tan x} dx = k \cos 4x + c$, then
(A) $k = -\frac{1}{2}$ (B) $k = -\frac{1}{8}$ (C) $k = -\frac{1}{4}$ (D) none of these
2. $\int \frac{e^x(1 + \sin x)}{1 + \cos x} dx$ is equal to
(A) $\log |\tan x| + c$ (B) $e^x \tan \left(\frac{x}{2} \right) + c$ (C) $\sin e^x \cot x + c$ (D) $e^x \cot x + c$
3. $\int \cos \sqrt{x} dx$ is equal to
(A) $2[\sqrt{x} \sin \sqrt{x} + \cos \sqrt{x}] + c$ (B) $\sin \sqrt{x} + c$
(C) $2[\sqrt{x} \cos \sqrt{x} - \sin \sqrt{x}] + c$ (D) none of these
4. $\int \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$ is equal to
(A) $\tan^{-1} \left(\frac{a}{b} \tan x \right) + c$ (B) $\frac{1}{ab} \tan^{-1} \left(\frac{b}{a} \cot x \right) + c$
(C) $\frac{1}{ab} \tan^{-1} \left(\frac{b}{a} \tan x \right) + c$ (D) $\tan^{-1} \left(\frac{b}{a} \tan x \right) + c$
5. $\int e^x \sec x(1 + \tan x) dx$ is equal to
(A) $e^x \sec x + c$ (B) $e^x \sec x \tan x + c$ (C) $e^x \tan x + c$ (D) none of these
6. $\int \left(x + \frac{1}{x} \right)^{3/2} \left(\frac{x^2 - 1}{x^2} \right) dx$ is equal to
(A) $\frac{5}{2} \left(x + \frac{1}{x} \right)^{5/2} + c$ (B) $\frac{2}{5} \left(x + \frac{1}{x} \right)^{5/2} + c$ (C) $2 \left(x + \frac{1}{x} \right)^{1/2} + c$ (D) none of these
7. $\int \frac{dx}{\sqrt{x+a} + \sqrt{x+b}} =$
(A) $\frac{2}{3} \cdot \frac{1}{(a-b)} [(x+a)^{3/2} - (x+b)^{3/2}] + c$ (B) $\frac{1}{2} \cdot \frac{1}{(a-b)} [(x+a)^{1/2} - (x+b)^{1/2}] + c$
(C) $\frac{3}{2} \cdot \frac{1}{(a-b)} [(x+a)^{3/2} + (x+b)^{3/2}] + c$ (D) none of these
8. $\int \frac{x^2 + 1}{\sqrt[3]{x^3 + 3x + 6}} dx =$
(A) $\frac{1}{2} (x^3 + 3x + 6)^{-1/2} + c$ (B) $-\frac{1}{2} (x^3 + 3x + 6)^{1/2} + c$
(C) $\frac{1}{2} (x^3 + 3x + 6)^{2/3} + c$ (D) none of these

9. $\int \sec^4 x \, dx =$
- (A) $\tan x + \frac{\tan^2 x}{3} + c$ (B) $\tan x + \frac{\tan^3 x}{3} + c$
(C) $\tan x + \frac{\tan^4 x}{3} + c$ (D) $\frac{\tan^4 x}{4} + c$
10. $\int_0^{\pi/2} \sin^6 \theta \cos^3 \theta \, d\theta =$
- (A) $\frac{2}{65}$ (B) $\frac{2}{63}$
(C) $\frac{1}{63}$ (D) $\frac{3}{130}$
11. If $\int \frac{\sqrt{\cot x}}{\sin x \cos x} dx = A \sqrt{\cot x} + B$, then $A =$
- (A) 1 (B) 2 (C) -1 (D) -2
12. If $\int \frac{(x^2 - 1)}{(x^4 + 3x^2 + 1) \tan^{-1}\left(\frac{x^2 + 1}{x}\right)} dx = k \log \left| \tan^{-1} \frac{x^2 + 1}{x} \right| + c$, then k is equal to
- (A) 1 (B) 2 (C) 3 (D) 5
13. $\int \frac{\cos 2x}{\cos x} dx$ is equal to
- (A) $2 \sin x + \log |(\sec x - \tan x)| + c$ (B) $2 \sin x - \log |(\sec x - \tan x)| + c$
(C) $2 \sin x + \log |(\sec x + \tan x)| + c$ (D) $2 \sin x - \log |(\sec x + \tan x)| + c$
14. $\int e^x \frac{1 + \sin x}{1 + \cos x} dx$ is
- (A) $\frac{e^x}{1 + \cos x} + c$ (B) $e^x \cot \frac{x}{2} + c$ (C) $e^x \tan \frac{x}{2} + c$ (D) None of these
15. $\int x^{13/2} \sqrt{1 + x^{5/2}} \, dx$ is equal to
- (A) $\frac{4}{5} \left[\frac{1}{7} (1 + x^{5/2})^{7/2} - \frac{2}{5} (1 + x^{5/2})^{5/2} + \frac{1}{3} (1 + x^{5/2})^{3/2} + c \right]$
(B) $\frac{4}{5} \left[\frac{1}{7} (1 + x^{5/2})^{7/2} - \frac{1}{5} (1 + x^{5/2})^{5/2} + (1 + x^{5/2})^{3/2} + c \right]$
(C) $\frac{4}{5} \left[(1 + x^{5/2})^{7/2} - \frac{2}{5} (1 + x^{5/2})^{5/2} + (1 + x^{5/2})^{3/2} + c \right]$
(D) none of these
16. If $\int f(x) \cos x \, dx = \frac{1}{2} f^2(x) + c$, then $f(x)$ can be
- (A) x (B) 1
(C) $\cos x$ (D) $\sin x$
17. The value of the integral $\int e^{\sin^2 x} (\cos x + \cos^3 x) \sin x \, dx$ is

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(A) $\frac{1}{2} e^{\sin^2 x} (3 - \sin^2 x) + c$

(B) $e^{\sin^2 x} \left(1 + \frac{1}{2} \cos^2 x \right) + c$

(C) $e^{\sin^2 x} (3 \cos^2 x + 2 \sin^2 x) + c$

(D) $e^{\sin^2 x} (2 \cos^2 x + 3 \sin^2 x) + c$

18. $\int \frac{dx}{\sqrt{2x-x^2}}$ is equal to

(A) $\sin^{-1} (1 - x) + c$

(B) $-\cos^{-1} (1 - x) + P$

(C) $\sin^{-1} (x - 1) + c$

(D) $\cos^{-1} (x - 1) + P$

19. $I = \int \frac{dx}{1+e^x}$ is equal to

(A) $\log_e \left(\frac{1+e^x}{e^x} \right) + c$

(B) $\log_e \left(\frac{e^x}{1+e^x} \right) + c$

(C) $\log_e (e^x) (e^x + 1) + c$

(D) $\log_e (e^{2x} + 1) + c$

20. $I = \int e^{\tan^{-1} x} \left(\frac{1+x^2+x}{1+x^2} \right) dx$ is equal to

(A) $xe^{\tan^{-1} x} + c$

(B) $x^2 e^{\tan^{-1} x} + c$

(C) $\frac{1}{x} e^{\tan^{-1} x} + c$

(D) none of these

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ANSWERS

1.	B	2.	B	3.	A	4.	C
5.	A	6.	B	7.	A	8.	C
9.	B	10.	A	11.	D	12.	A
13.	D	14.	C	15.	A	16.	D
17.		18.	C	19.	B	20.	A

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